

8(a)(i)	The relative speed of approach of the two objects is greater than their relative speed of separation. The relative speed of separation in the case of a perfectly inelastic collision is zero.
(a)(ii)	The total kinetic of the two objects before the collision is greater than the total kinetic energy of the two objects after the collision, as total kinetic energy is not conserved and converted into other forms of energy.
(b)	Gain in K.E. = loss in G.P.E. $\frac{1}{2}mv^2 = mgL(1 - \cos 21^\circ)$ $v^2 = 2gL(1 - \cos 21^\circ)$ $v^2 = 2(9.81)(1.50)(1 - \cos 21^\circ)$ $v = 1.398 \text{ m s}^{-1}$ $v \approx 1.4 \text{ m s}^{-1}$
(c)(i)	By conservation of linear momentum, $m_A u_A + m_B u_B = m_A v_A + m_B v_B$ $(0.096)(1.4) + 0 = (0.096)(-0.79) + p_B$ $p_B = 0.21 \text{ N s}$

(c)(ii)	K.E. lost by A $= \frac{1}{2} m_A u_A^2 - \frac{1}{2} m_A v_A^2$ $= \frac{1}{2} (0.096)(1.4)^2 - \frac{1}{2} (0.096)(0.79)^2$ $= 0.06412 \text{ J}$ K.E. lost in this collision $= 0.014 \times 0.09408$ $= 0.01317 \text{ J}$ K.E. of ball B $= 0.06412 - 0.01317$ $= 0.051 \text{ J}$
(d)(i)	$KE_B = \frac{1}{2} m_B v_B^2$ $p_B = m_B v_B$ $\therefore v_B = \frac{2KE_B}{p_B} = \frac{2(0.051)}{0.21} = 0.49 \text{ m s}^{-1}$
(d)(ii)	$m_B = \frac{p_B}{v_B} = \frac{0.21}{0.486} = 0.43 \text{ kg}$
(e)	Both spheres cannot be stationary at the same time because the initial momentum of the system was not zero but equal to the initial momentum of A (0.134 N s) before the collision. Since no net external force acts on the system, the total momentum of the system must remain constant at 0.134 N s at all times. If both are stationary at the same instant, the total momentum of the system will be zero at that instant and this will violate the principle of conservation of linear momentum.
(f)	Magnitude of impulse of sphere A = magnitude of change in momentum of sphere A $= m_A v_A - m_A u_A $ $= (0.096)(-0.79) - (0.096)(1.4) $ $= 0.21 \text{ N s}$

(g)	The impulse on A is horizontal and to the left. Since no external force acts, by the principle of conservation of linear momentum, the impulse on B must be horizontal and to the right. This is only possible if the contact force at the point of collision between the two bodies is horizontal. Since both are spheres and contact force act along the line joining their centres of spheres, in order for contact force to be horizontal, the centre of spheres must be on the same horizontal level at the time of collision.
9(a)(i)	A field of force is a region of space where a body possessing a particular property